

PYTHON PROGRAMMING LAB MANUAL

Week 2 — Introduction to Python and Basic Constructs

Program	Mechanical Engineering	Semester	IV
Course Name	Programming Concepts in Mechanical Engineering	Course Code	25ME34I
Week	2	Duration	3 Hours (Practice Session)
Course Outcome	CO-1: Use appropriate syntaxes while writing program.	Program Outcomes	PO-1, PO-2

Learning Objectives

- Get comfortable with Python syntax, variables, and basic data types (int, float, string, boolean).
- Use input() and print() to build simple interactive programs.
- Apply type conversion functions (int(), float(), str()) correctly.
- Use arithmetic and assignment operators to solve small engineering calculations.

Quick Theory Recap

- Python environment: code can be written and run in Anaconda, Jupyter Notebook, or an IDE such as PyCharm.
- Data types: int (whole numbers), float (decimals), string (text), boolean (True / False).
- input() always returns a string — convert it with int() or float() before doing arithmetic.
- print() displays output; multiple values can be separated by commas.
- Arithmetic operators: + - * / // % **. Assignment operators: = += -= *= /=.

Practical Exercises

Exercise 1: Basic Arithmetic Operations

Aim: Write a Python program to accept two numbers from the user and perform addition, subtraction, multiplication, and division.

Program:

```
# Program to perform arithmetic operations

# Accept two numbers from the user
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

# Arithmetic operations
addition = a + b
subtraction = a - b
```

```

multiplication = a * b

# Check to avoid division by zero
if b != 0:
    division = a / b
else:
    division = "Not possible (division by zero)"

# Display results
print("Addition:", addition)
print("Subtraction:", subtraction)
print("Multiplication:", multiplication)
print("Division:", division)

```

Sample Output

```

Enter first number: 10
Enter second number: 4
Addition: 14.0
Subtraction: 6.0
Multiplication: 40.0
Division: 2.5

```

Explanation: input() collects text from the keyboard; float() converts it to a decimal number so arithmetic can be performed. An if-else check prevents a crash if the user enters 0 as the second number.

Exercise 2: Output Speed and Torque in a Gear System

Aim: Calculate output speed and output torque of a gear system from user-entered input speed, input torque, and gear ratio.

Concept:

- Gear Ratio (GR) = Input Speed ÷ Output Speed
- Output Speed = Input Speed ÷ Gear Ratio
- Output Torque = Input Torque × Gear Ratio

Program:

```

# Program to calculate output speed and torque in a gear system

# Input values from user
input_speed = float(input("Enter input speed (RPM): "))
input_torque = float(input("Enter input torque (Nm): "))
gear_ratio = float(input("Enter gear ratio: "))

# Calculations using basic operators
output_speed = input_speed / gear_ratio
output_torque = input_torque * gear_ratio

# Display results
print("Output Speed (RPM):", output_speed)

```

```
print("Output Torque (Nm):", output_torque)
```

Sample Output

```
Enter input speed (RPM): 1200
Enter input torque (Nm): 50
Enter gear ratio: 4
Output Speed (RPM): 300.0
Output Torque (Nm): 200.0
```

*Explanation: This is a real mechanical-engineering use of arithmetic operators — the same / and * used above, applied to a gearbox calculation.*

Exercise 3: Working with Variables

Aim: Store values of different data types in variables and use them in expressions.

3(a) Name, Age, Height

```
name = "Rahul"
age = 18
height = 5.6
print(name, age, height)
```

Sample Output

```
Rahul 18 5.6
```

Explanation: name is a string, age is an int, height is a float — Python variables do not need a declared type.

3(b) Sum and Product

```
a = 5
b = 10
print(a + b)
print(a * b)
```

Sample Output

```
15
50
```

*Explanation: + adds the two values; * multiplies them.*

3(c) Area of a Circle

```
pi = 3.14
r = 7
```

```
area = pi * r * r
print(area)
```

Sample Output

```
153.86
```

Explanation: Implements the formula $Area = \pi r^2$.

Exercise 4: Boolean Data Type

Aim: Demonstrate the boolean data type and its use in engineering decision-making.

Program:

```
# Boolean example: checking if a machine is running
machine_running = True
temperature_ok = True

# A boolean expression combining two conditions
safe_to_operate = machine_running and temperature_ok

print("Machine running:", machine_running)
print("Temperature OK:", temperature_ok)
print("Safe to operate:", safe_to_operate)
```

Sample Output

```
Machine running: True
Temperature OK: True
Safe to operate: True
```

Explanation: A boolean variable holds only True or False. In engineering programs, booleans are used to represent machine/sensor status and to combine multiple safety conditions before an action is taken.

Exercise 5: Type Conversion

Aim: Convert between data types using `int()`, `float()`, and `str()`, and observe the results.

5(a) Integer + Float

```
a = 5
b = 2.5
c = a + b
print(c, type(c))
```

Sample Output

```
7.5 <class 'float'>
```

Explanation: Python automatically (implicitly) converts the int to a float so the addition can be performed.

5(b) String to Integer

```
a = "25"
b = int(a)
print(b + 5)
```

Sample Output

```
30
```

Explanation: A string coming from input() must be explicitly converted with int() before it can be used in arithmetic.

5(c) Float to Integer

```
x = int(98.6)
print(x)
```

Sample Output

```
98
```

Explanation: int() truncates (removes) the decimal part — it does not round the number.

Exercise 6: Bonus: Input-Based Decision Problems (Preview of Week 3)

Note: These two problems use the if-else statement, which is formally covered in Week 3 (Control Structures). They are included here only as an optional extension for students who finish early — mastering input() and type conversion first is the priority for this week.

6(a) Pass / Fail

```
marks = int(input())

if marks >= 35:
    print("Pass")
else:
    print("Fail")
```

Sample Output

```
40
Pass
```

6(b) Voting Eligibility

```
age = int(input())
```

```
if age >= 18:  
    print("Eligible")  
else:  
    print("Not Eligible")
```

Sample Output

```
20  
Eligible
```

Self-Practice (To Be Completed Individually)

Attempt the following on your own and get your output verified by the faculty:

- Write a Python program that accepts the length and breadth of a rectangular plate and prints its area and perimeter.
- Write a Python program that converts a temperature entered in Celsius to Fahrenheit ($F = C \times 9/5 + 32$).
- Write a Python program that stores a boolean value indicating whether a pressure vessel has passed a safety test, and prints an appropriate message.
- Write a Python program that accepts a value as a string, converts it to a float, and prints its data type before and after conversion.

Reference: Week 2 syllabus content (25ME34I — Programming Concepts in Mechanical Engineering, DTE Karnataka).